Statistical Foundations of Machine Learning

Practice Set 2

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1 Homework problems

Note: solutions to problems in this section are to be submitted and will be graded. Deadline for submission is Sunday, 20th Sep.. Total marks : 15. Justify your answers and present all calculations/arguments clearly.

1. **(Feature rollout)**

   A company rolls out a new feature to users. Each user independently receives the feature with probability $1/2$. Suppose 5 users log in.

   (a) (1 mark) Find the probability that *exactly* 3 out of 5 users receive the feature. Write down both the formula and the computed value.

   (b) (1 mark) Find the probability that *at least* 4 users receive the feature. Write down both the formula and the computed value.

   (c) (1 mark) You are now given that not all 5 users receive the features, i.e., *at least one of them has not received the feature.* Find the *conditional probability* that exactly 3 users receive it.

   Hint: You might find the binomial distribution useful.

2. Teams A and B play a 'best-of-five' series with the first team to win three games wins the series (the series ends immediately when such an event happens and no further games are played then). A wins a game with probability p independently of the result of the other games.

   Let X_k be the indicator variable for A winning the series in k games. Note that for any event $\mathcal{E}$, the indicator random variable $1_{\mathcal{E}}$ takes two values: 1 if $\mathcal{E}$ is true (i.e. with probability $\mathbb{P}(\mathcal{E})$) and 0 otherwise.

   (a) (2 marks) For X_k, determine its probability mass function (pmf) as a function of p.

   (b) (2 marks) Let X be the indicator variable that A wins the series. Determine the pmf of X.

3. A random variable X is said to have the $\mathrm{Laplace}(a, b)$ distribution if it has the following pdf with constants $b > 0$, a, and C.

$$f_X(x) = C \ \exp\left(-b|x - a|\right), \quad x \in \mathbb{R}.$$

   (a) (1 mark) Plot the function for, say, $a = 1, b = 2$.

(b) (1 mark) Determine C in terms of a and b that would make this a valid density function.

(c) (2 marks) Find the cumulative distribution function (CDF) of X.

4. (4 marks) You are stuck in a cave, with three tunnels in front of you. The first tunnel takes you out of the cave in 3 hours. The second tunnel brings you back to the starting position in 5 hours. The third tunnel also brings you back, but in 7 hours. Entrances to the tunnels are identical. What is the expected time for you to get out of the cave ?

2 Practice problems

Note: problems in this section are for practice, solutions not to be submitted

1. Let X be a discrete random variable with the following PMF -

$$P_X(x) = \begin{cases} 0.1 & x = 1, 2, 10 \\ 0.3 & x = 5 \\ 0.4 & x = 8 \\ 0 & \text{otherwise} \end{cases}$$

Find and plot the CDF of X.

2. Buses arrive at a bus-stop at 10 minute intervals starting at noon. A man arrives at the bus-stop at a random time X minutes after noon, where X has the CDF:

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{60} & 0 \leq x \leq 60 \\ 1 & x > 60 \end{cases}.$$

What is the probability that the man waits less than 5 minutes for a bus?

3. I roll a fair die repeatedly until a number larger than 4 is observed. If N is the total number of times that I roll the die, find $P(N = k)$, for $k = 1, 2, 3, \cdots$.

4. The random variable X has the following CDF.

$$F_X(x) = \begin{cases} x & \text{for } 0 \leq x < 1/3 \\ \frac{3x+1}{6} & \text{for } 1/3 \leq x < 2/3 \\ x & \text{for } 2/3 \leq x < 1 \\ 1 & \text{for } x \geq 1 \end{cases}$$

(a) What is $\mathbb{P}(X < 1/2)$?

(b) What is $\mathbb{P}(X < 2/3)$?

(c) Is X a continuous random variable?

5. **CDF & PDF** The probability density function (PDF) of a random variable has the following form.

$$f(x) = \begin{cases} a & \text{for } x < 0 \\ x & \text{for } x \in [0,1] \\ b(1-x) & \text{for } x \in (1,2] \\ c & \text{for } x > 2 \end{cases}$$

Here, a, b, c are constants.

   (a) What are the values of a, b and c?

   (b) Compute the CDF corresponding to this PDF.

6. A positive integer is selected, with $P(I = n) = \frac{1}{2^n}, n = 1, 2, \cdots$. If I takes the value n, a coin with probability of heads equal to e^{-n} is tossed once. Find the probability that the resulting toss will be a head.

7. X is a strictly positive discrete random variable, i.e., it takes values in the set $\{1, 2, \ldots, \}$ and $p_{\mathsf{X}}(x)$ is its pmf. If the pmf has the form $p_{\mathsf{X}}(x) = \frac{C*2^x}{x!}$, find the value of C. Repeat for $p_{\mathsf{X}}(x) = \frac{C*2^{-x}}{x}$, and $p_{\mathsf{X}}(x) = Cx^{\alpha}$.

8. For the previous problem, find (i) $\mathsf{Prob}(\mathsf{X} > 2)$, (ii) most probable value of X.

9. There are n coins each of which comes up heads with probability p. In round 1, each coin is tossed once. In round 2, all those coins that came up heads in round 1 are tossed again and others are left unchanged. Let X be the total number of heads after round 2. Find the pmf of X.

10. Let X be the random variable denoting the number of tosses of a biased coin to get the first head. The first head has not occurred in the first k tosses. What is the probability that you will have to toss the coin n more times.